

This analysis uses weekly stock data for Nike Inc. and Apple Inc. The datasets include different variables, calculated weekly returns, assets relationship, and the regression analysis, which allow us to observe how the performance of each stock changes over time and assess their investment characteristics.

The summary statistics provide information concerning the major aspects of the returns for the two companies. Nike has an average weekly return of approximately -0.004 , while Apple has a slightly positive mean return of 0.0025 . The standard deviation, which provides information concerning the risks, is 0.0548 for Nike and 0.0455 for Apple. These figures indicate that the returns for Nike are more volatile than the returns for Apple. Additionally, the minimum and maximum returns also demonstrate that both stocks experience significant weekly variations, confirming that stock investments involve uncertainty and risk.

The correlation coefficient for the returns of the two stocks is approximately 0.34 , which indicates that the two stocks have a moderate positive relationship. However, the relationship is not too strong, and investing in the two stocks may help in the diversification of the portfolio and the reduction of the overall risks.

The regression analysis is concerned with how the individual company's stock return is dependent on the overall market return. Separate regression analyses were done for Nike and Apple, using the market return as the independent variable. The regression analysis for Nike's stock return found that it explained 37% of the variation in the company's stock return. The beta coefficient of 2.36 means that the company's stock is highly sensitive to market fluctuations. For every 1% change in market return, there is a corresponding change of 2.36% in the company's stock return. The analysis found that the company's stock return is heavily dependent on market fluctuations.

As can be seen, only 10% of the variation can be explained by the change in the market return. The beta coefficient of 1.21 indicates that the stock of Apple tends to follow the market in the same direction and is more sensitive to changes in the market return. However, the statistical results indicate that the relationship is not statistically significant. The results indicate that although the returns of Apple have a positive relationship with the market return, the market return does not significantly explain the changes in the return of Apple.

A chi-square test was conducted to evaluate whether the population variance of the stock returns differs from the hypothesized variance. The test used 53 observations, giving 52 degrees of freedom, and the calculation was based on the sample variance obtained from the weekly returns. I used a chi-square test to check whether the annual standard deviation of the stock returns differs from 0.25 (25%), based on the sample variance of weekly returns. An F-test was also performed to compare the variances of Apple and Meta in order to determine whether their levels of volatility differ. The test used the calculated sample variances of 0.00207 for Apple and 0.00253 for Meta. The comparison shows that Meta has a larger variance, indicating higher variability in weekly returns compared to Apple.

